Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/14/2006
PI Name:	Goree, John Ph.D.		
Project Title:	Self-Structuring in Dusty Plasmas		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john-goree@uiowa.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	319-335-1843
Organization Name:	University of Iowa		
PI Address 1:	Department of Physics and Astronomy		
PI Address 2:	210 Van Allen Hall		
PI Web Page:			
City:	Iowa City	**State:**	IA
Zip Code:	52242-1403	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	01/01/2005	**End Date:**	03/31/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Rogers, Richard	**Contact Phone:**	216-433-6512
Contact Email:	Richard.B.Rogers@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2007 (from 5/18/09), per info from PI stating that grant was reinstated as a new grant ; see "Self-Structuring in Dusty Plasmas (NNX07AD22G)" for new information (10/2007) NOTE: grant reinstated 8/2006; end date now 5/18/2009, per J. Cohen (HQ)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC05GA43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Experiments are performed to study self-structuring of a dusty plasma due to weak forces. A dusty plasma consists of particles of solid matter suspended in an ionized gas. Micron-size particles acquire a large negative charge of typically 10,000 elementary charges. Due to inter-particle repulsion, they arrange themselves in structure with solid or liquid-like properties, similar to a colloidal crystal. With particles of this size, gravity is a dominant force, causing particles to sediment to a thin layer at the bottom of the plasma. In this project, the collective behavior of particles in a dusty plasma are explored. Work includes ground-based experiments to prepare for future flight experiments using ESA facilities on ISS. This project represents the PI's role in a collaboration with German co-I's: Professor Alexander Piel of the University of Kiel, and Professor Andre Melzer of the University of Greifswald.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During this one-year funding period: * we finished up some previous projects by publishing papers * we carried out two new experimental campaigns of laboratory experiments * we began developing hardware for the next campaign. These three campaigns are summarized below: The first campaign had two goals for the study of a 2D liquid. o We demonstrated a new method of heating a dusty plasma using rapidly-moving powerful laser beams to manipulate particles. This is significant because it enables new classes of experiments to study collective effects and transport processes, which in general depend on temperature and temperature gradients. This method involves a complicated laser-manipulation setup because our sample consists of electrically-charged microspheres suspended in space that cannot simply be placed in thermal contact with a solid heatsink. Having this possibility of varying the sample temperature and temperature gradients enables experiments that were previously impractical for dusty plasmas and for colloidal suspensions as well. We have now published this new method in Physics of Plasmas. o We made the first experimental observation of the cutoff wavenumber for shear waves in a liquid strongly-coupled plasma. Seismologists have long exploited the fact that shear waves do not easily propagate through liquids as a way of detecting that the Earth's core is a liquid. It has also long been known that molecular liquids, such as molten metals, can sustain shear wave propagation only for very short wavelengths, i.e., wave numbers above a "cutoff." The existence of such a cutoff has also been predicted in many theoretical papers to occur in all types of strongly-coupled plasmas, but it was never demonstrated experimentally until now, in our experiment. Using our new laser-heating method, we varied the temperature of our sample, and we measured the phonon spectrum for various temperatures. The phonon spectrum revealed the presence of the cutoff. This work has been submitted to PRL. The results for this first campaign encourage us to carry out future flight experiments on ISS, using the IMPF facility, which will be equipped with a manipulation laser. Experiments on IMPF will differ from our lab experiments because they will be 3D rather than 2D. The second campaign was an investigation of nonlinear pulses in a 2 D crystal. We used the manipulation laser to apply a powerful pulse to excite nonlinear compressional waves. We searched for evidence of a theoretically-predicted rarefactive wave in a 2D triangular lattice with a long-range repulsive potential. Our analysis of our data indicates that no such rarefactive wave occurs. As we continue our analysis, if our conclusions remain the same, we will write a paper intended to disprove the theoretical prediction. Next campaign: We are setting up a campaign of experiments to search for experimental indications of superdiffusion or diffusion in 2D liquids. There is presently a controversy among theorists regarding whether 2D liquids can ever exhibit ordinary diffusion, as is common in 3D liquids, and we intend these experiments to help resolve this controversy. We will use direct imaging of particle motion to determine whether motion is diffusive or superdiffusive in our laser-heated 2D liquid dusty plasmas. This next campaign will serve as a preparation for future flight experiments on ISS, using the IMPF facility which will be equipped with a manipulation laser and cameras similar to those used in our lab experiments. The lab experiments are 2D, whereas the flight experiments will use a 3D suspension. Collaboration with German co-I's This project is intrinsically international. It is intended to prepare for future flight experiments using ESA facilities on the ISS. Twice during 2005 the PI hosted German co-I, Professor Alexander Piel of the University of Kiel, who participated in our first campaign of experiments, as described above, during a month-long visit in May 2005. In October 2005 he visited again, and we began writing three papers. Two of these papers have been published by Physics of Plasmas, and one more was recently submitted to Physical Review Letters. During 2006, the PI plans to host a postdoc from Piel's group for a period of three months, to carry out our next campaign of laboratory experiments. The PI is in the process of co-authoring a book chapter written jointly with the other German co-I, Professor Andre Melzer of the University of Greifswald.
Bibliography Type:	Description: (Last Updated: 01/23/2025)
Abstracts for Journals and Proceedings	Nosenko V. , Bin Liu and J. Goree. "Shear viscosity of a liquid strongly-coupled dusty plasma." International Conference on Dusty Plasmas, Orleans, France, June 2005. Presented at the International Conference on Dusty Plasmas, Orleans, France, June 2005. , Jun-2005

Abstracts for Journals and Proceedings	Goree J., Z. Donko, P. Hartmann and K. Kutasi. "Viscosity of strongly-coupled dusty plasmas in a liquid state." 47th Annual Meeting of the APS Division of Plasma Physics, Denver, CO, October 2005. Presented at the 47th Annual Meeting of the APS Division of Plasma Physics, Oct 2005. , Oct-2005
Abstracts for Journals and Proceedings	Goree J. "Waves in Dusty Plasmas." Invited Talk at the International Conference on Dusty Plasmas, Orleans, France, June 2005. International Conference on Dusty Plasmas, June 2005. , Jun-2005
Abstracts for Journals and Proceedings	Goree J. "Wave Experiments in Dusty Plasmas: Linear and Nonlinear. " Invited talk at the APS April 2006 Meeting mini-conference on Laboratory Experiments on Plasma Astrophysics. APS April Meeting mini-conference on Laboratory Experiments on Plasma Astrophysics, April 2006. , Apr-2006
Articles in Peer-reviewed Journals	Piel A, Nosenko V, Goree J. "Laser-excited shear waves in solid and liquid two-dimensional dusty plasmas." Phys Plasmas. 2006 Apr;13(4):042104. , Apr-2006
Articles in Peer-reviewed Journals	Liu B, Goree J. "Shear viscosity of two-dimensional Yukawa systems in the liquid state." Phys Rev Lett. 2005 May 13;94(18):185002. PMID: 15904380 , May-2005
Articles in Peer-reviewed Journals	Liu B, Goree J, Vaulina OS. "Test of the Stokes-Einstein relation in a two-dimensional Yukawa liquid." Phys Rev Lett. 2006 Jan 13;96(1):015005. PMID: 16486470 , Jan-2006
Articles in Peer-reviewed Journals	Nosenko V, Goree J, Skiff F. "Bispectral analysis of nonlinear compressional waves in a two-dimensional dusty plasma crystal." Phys Rev E Stat Nonlin Soft Matter Phys. 2006 Jan;73(1 Pt 2):016401. PMID: 16486282 , Jan-2006
Articles in Peer-reviewed Journals	Donkó Z, Goree J, Hartmann P, Kutasi K. "Shear viscosity and shear thinning in two-dimensional Yukawa liquids." Phys Rev Lett. 2006 Apr;96(14):145003. , Apr-2006
Articles in Peer-reviewed Journals	Nosenko V, Goree J, Piel A. "Laser method of heating monolayer dusty plasmas." Phys Plasmas 2006 Mar;13(3):032105 , Mar-2006